Coordinates and Tables of Values

Name _____ Date _____ No calculator Version 1

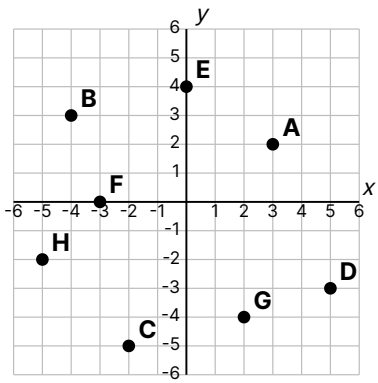
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KEY WORDS coordinates *(taunga)* · x-axis *(tuaka-x)* · y-axis *(tuaka-y)* · table of values *(tūtohi uara)*

A Coordinates in four quadrants ●○○

Remember: The first number is how far across (x) and the second is how far up or down (y). The point $(3, -2)$ is 3 right and 2 down.

1. Write the coordinates of each point.



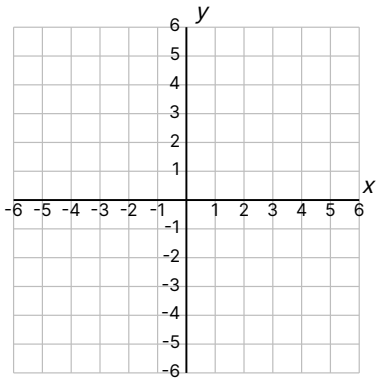
a) A _____ **b)** B _____

c) C _____ **d)** D _____

e) E _____ **f)** F _____

g) G _____ **h)** H _____

2. Plot these points on the grid and label each one with its letter: P $(4,\ 1)$, Q$(-3,\ 2)$, R$(-2,\ -4)$, S$(5,\ -3)$, T$(0,\ 3)$, U$(-5,\ 0)$.



B Quadrants ●○○

Remember: The axes cut the grid into four quadrants. The first is top right (x and y both positive), and they are numbered anticlockwise: second top left, third bottom left, fourth bottom right.

3. Which quadrant is each point in?

a) $(3,\ 5)$ _____ **b)** $(-2,\ 4)$ _____ **c)** $(4,\ -1)$ _____

d) $(-6,\ -3)$ _____ **e)** $(-1,\ 7)$ _____ **f)** $(8,\ -8)$ _____

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C Tables of values ●●○

Remember: Put each value of x into the rule to find y. For $y = 2x + 1$ and $x = 3$: $y = 2 \times 3 + 1 = 7$.

4. Complete the table of values for each rule.

a) $y = x + 3$

x	0	1	2	3	4
y					

b) $y = 2x$

x	0	1	2	3	4
y					

c) $y = 2x + 1$

x	-2	-1	0	1	2
y					

d) $y = 3x - 2$

x	-1	0	1	2	3
y					

e) $y = -x + 4$

x	-2	-1	0	1	2
y					

f) $y = -2x + 1$

x	-1	0	1	2	3
y					

D Finding the rule ●●●

Remember: Look at how much y changes each time x goes up by 1. That number multiplies x. Then look at the value of y when $x = 0$.

5. Write the rule for each table in the form $y = \ldots$

a)

x	0	1	2	3	4
y	5	6	7	8	9

Rule: _____

b)

x	0	1	2	3	4
y	0	3	6	9	12

Rule: _____

c)

x	0	1	2	3	4
y	3	5	7	9	11

Rule: _____

d)

x	0	1	2	3	4
y	-1	3	7	11	15

Rule: _____

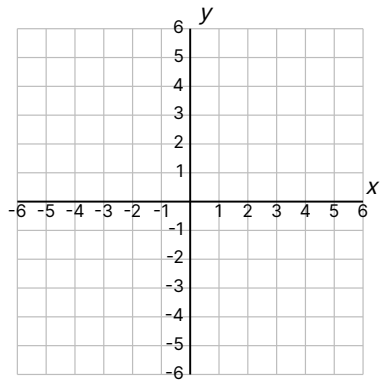
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E Plotting a rule ●●●

6. Complete each table, then plot its points on the grid.

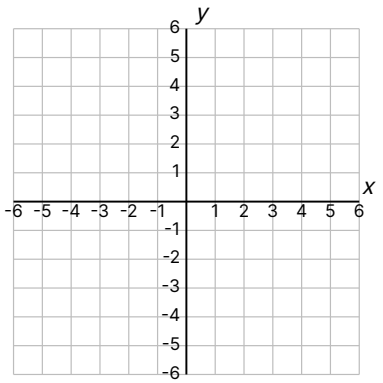
a) $y = x + 2$

x	-2	-1	0	1	2
y					



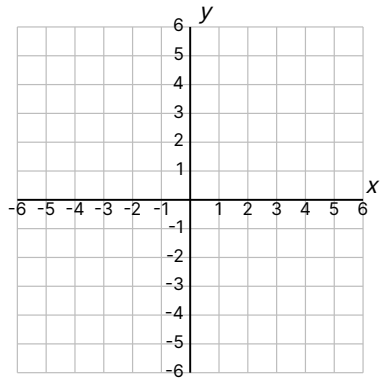
b) $y = 2x - 1$

x	-2	-1	0	1	2
y					



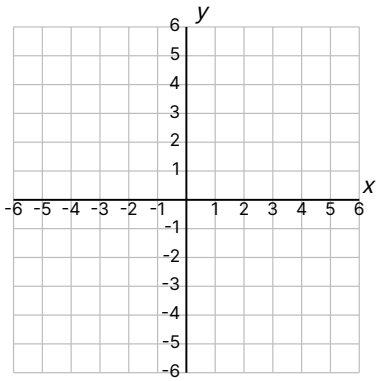
c) $y = -x + 3$

x	-2	-1	0	1	2
y					



d) $y = 3x - 2$

x	-1	0	1	2
y				



7. Look at the points you plotted on each grid. What do you notice about them?

Extension

8. Does the point fit the rule? Show your working.

a) $(4,\ 9)$ and $y = 2x + 1$ **b)** $(5,\ 12)$ and $y = 3x - 2$ **c)** $(-2,\ 8)$ and $y = -x + 6$

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Coordinates and Tables of Values

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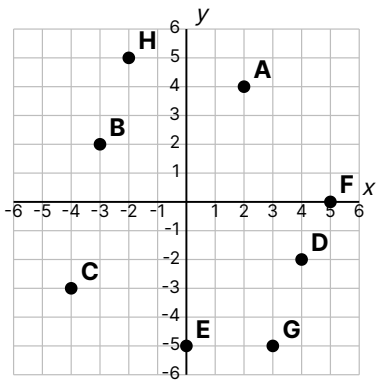
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KEY WORDS coordinates *(taunga)* · x-axis *(tuaka-x)* · y-axis *(tuaka-y)* · table of values *(tūtohi uara)*

A Coordinates in four quadrants ●○○

Remember: The first number is how far across (x) and the second is how far up or down (y). The point $(3, -2)$ is 3 right and 2 down.

1. Write the coordinates of each point.



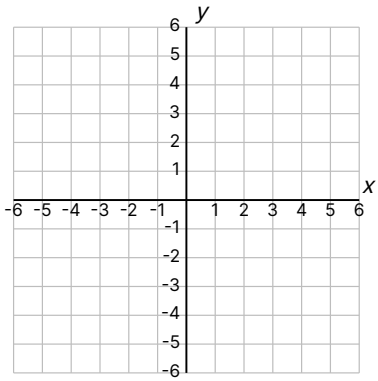
a) A _____ **b)** B _____

c) C _____ **d)** D _____

e) E _____ **f)** F _____

g) G _____ **h)** H _____

2. Plot these points on the grid and label each one with its letter: P $(3,\ 2)$, Q$(-4,\ 1)$, R$(-3,\ -5)$, S$(2,\ -4)$, T$(0,\ -2)$, U$(4,\ 0)$.



B Quadrants ●○○

Remember: The axes cut the grid into four quadrants. The first is top right (x and y both positive), and they are numbered anticlockwise: second top left, third bottom left, fourth bottom right.

3. Which quadrant is each point in?

a) $(4,\ 2)$ _____ **b)** $(-5,\ 1)$ _____ **c)** $(2,\ -6)$ _____

d) $(-3,\ -4)$ _____ **e)** $(-7,\ 2)$ _____ **f)** $(5,\ -5)$ _____

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C Tables of values ●●○

Remember: Put each value of x into the rule to find y. For $y = 2x + 1$ and $x = 3$: $y = 2 \times 3 + 1 = 7$.

4. Complete the table of values for each rule.

a) $y = x + 2$

x	0	1	2	3	4
y					

b) $y = 3x$

x	0	1	2	3	4
y					

c) $y = 2x + 3$

x	-2	-1	0	1	2
y					

d) $y = 3x - 1$

x	-1	0	1	2	3
y					

e) $y = -x + 5$

x	-2	-1	0	1	2
y					

f) $y = -2x + 3$

x	-1	0	1	2	3
y					

D Finding the rule ●●●

Remember: Look at how much y changes each time x goes up by 1. That number multiplies x. Then look at the value of y when $x = 0$.

5. Write the rule for each table in the form $y = \ldots$

a)

x	0	1	2	3	4
y	4	5	6	7	8

Rule: _____

b)

x	0	1	2	3	4
y	0	4	8	12	16

Rule: _____

c)

x	0	1	2	3	4
y	5	7	9	11	13

Rule: _____

d)

x	0	1	2	3	4
y	-1	2	5	8	11

Rule: _____

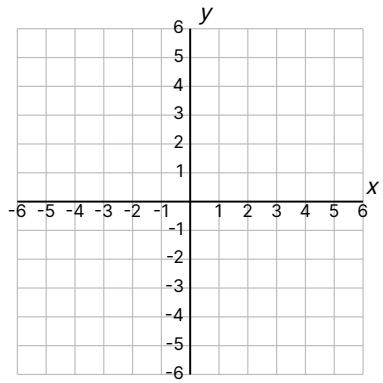
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E Plotting a rule ●●●

6. Complete each table, then plot its points on the grid.

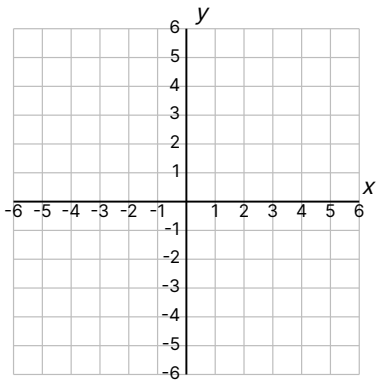
a) $y = x + 1$

x	-2	-1	0	1	2
y					



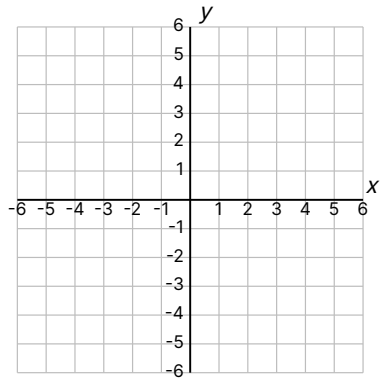
b) $y = 2x - 3$

x	-1	0	1	2	3
y					



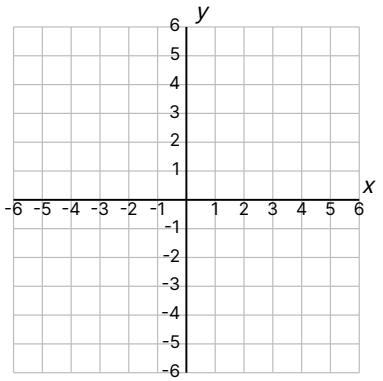
c) $y = -x + 4$

x	-2	-1	0	1	2
y					



d) $y = 3x - 1$

x	-1	0	1	2
y				



7. Look at the points you plotted on each grid. What do you notice about them?

Extension

8. Does the point fit the rule? Show your working.

a) $(4,\ 11)$ and $y = 2x + 3$ **b)** $(5,\ 15)$ and $y = 3x - 1$ **c)** $(-3,\ 8)$ and $y = -x + 5$

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